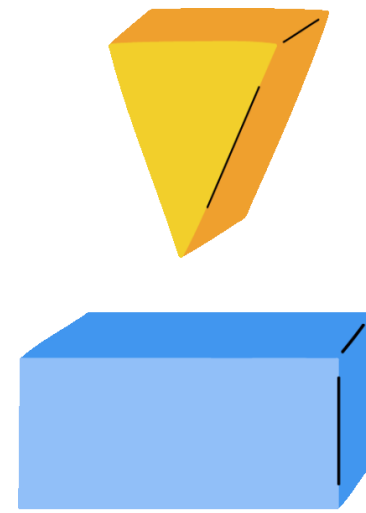
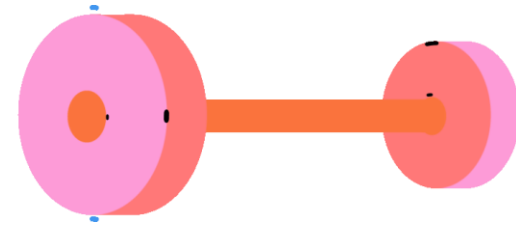
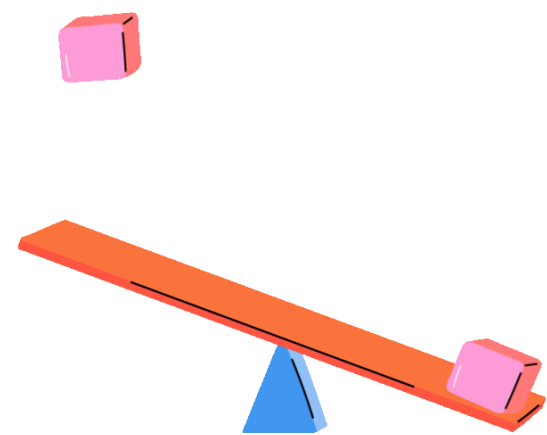


# SIMPLE AND COMPOUND MACHINES



# LEARNING OBJECTIVES:

At the end of the lesson, the students must be able to:

- Explain the characteristics of efficient simple and compound machines.
- Apply concepts of translational and rotational motion to design and build prototypes of efficient simple and compound machines



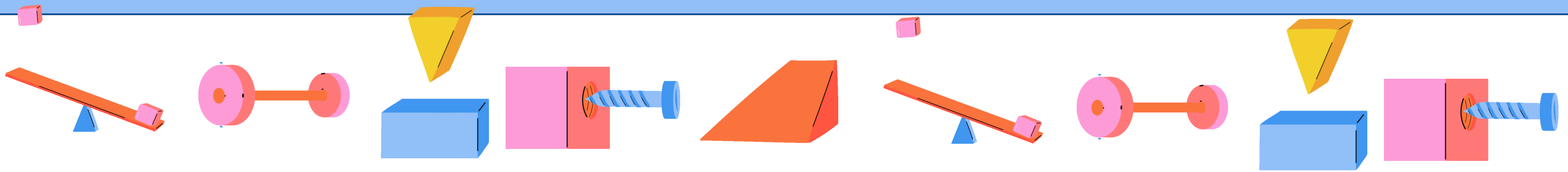
# TOPIC OUTLINE:

I. Simple and  
Compound Machines

II. Characteristics of  
Efficient Machines



# SIMPLE MACHINES



# SIMPLE MACHINES

- Most **basic device** that **performs work**
- Exerts a **single force** on a single load (**push or pull**)
- Foundational element in the creation of more complex machines

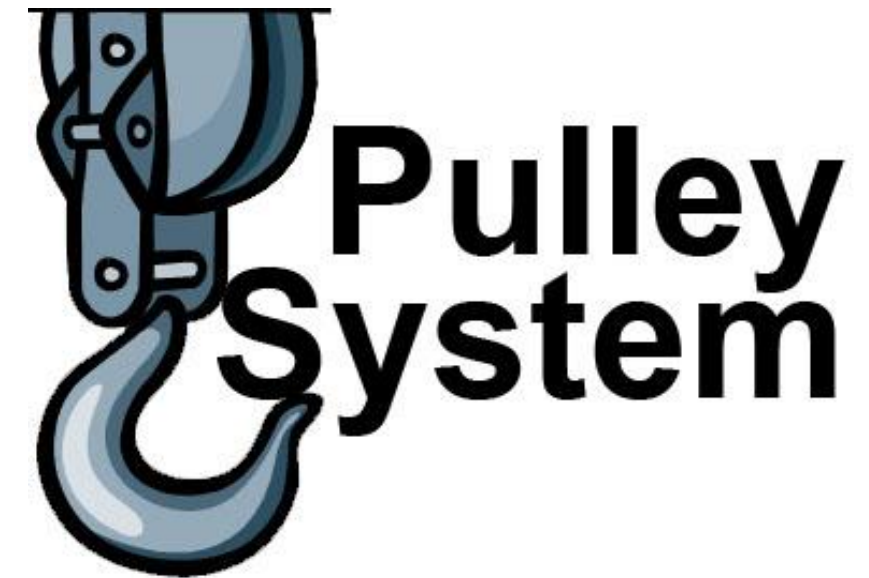
# SIMPLE MACHINES

- A device that allows work to be performed with less effort.
- A device with few or no moving parts.
- A device that makes work easier by changing force, distance, or by changing the direction of the force.

# SIMPLE MACHINES

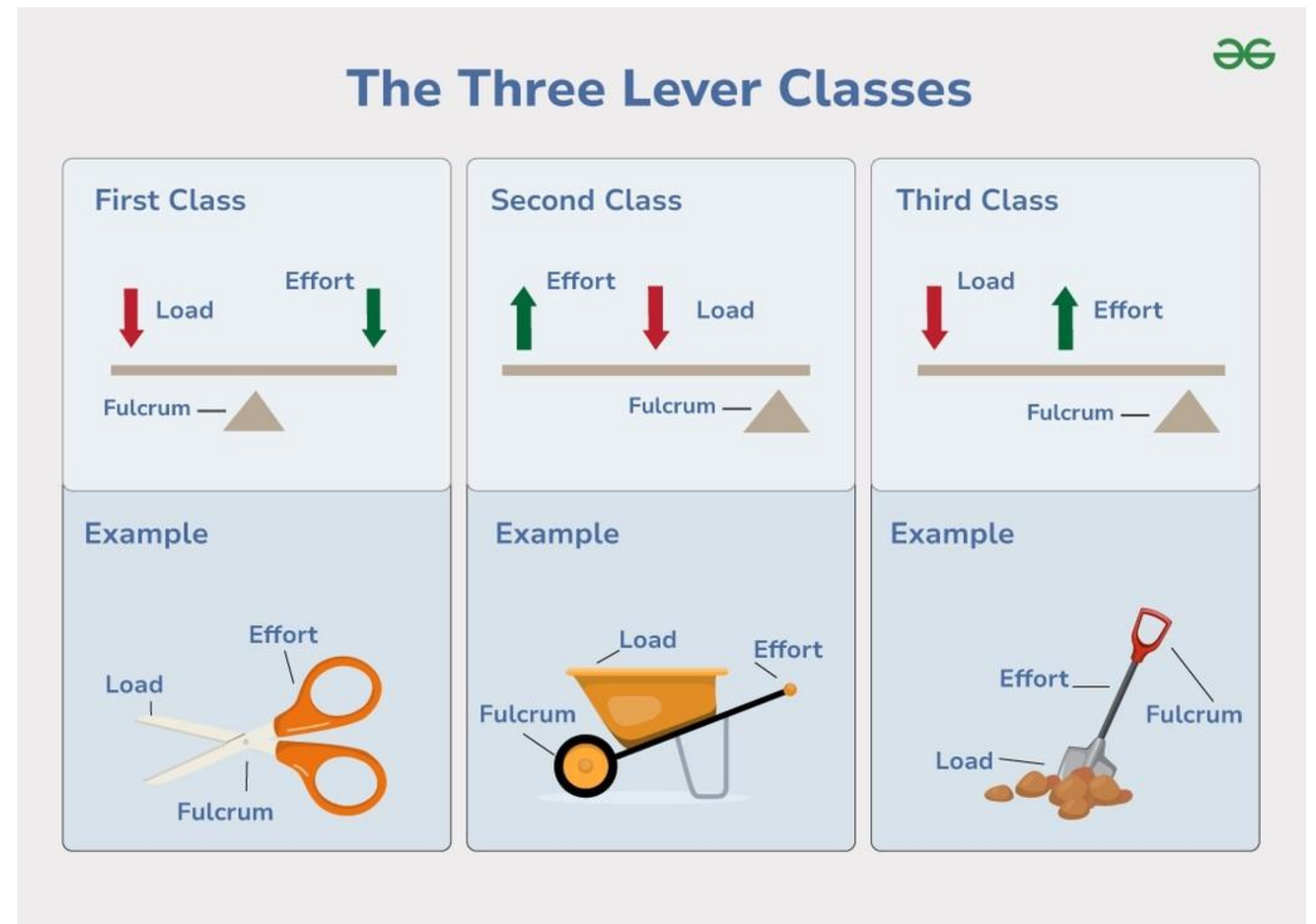
- Does not have an energy source
- Cannot accomplish more work than the amount of force applied
- **Can change direction or distribution of force** to make work more manageable

# The 6 Simple Machines



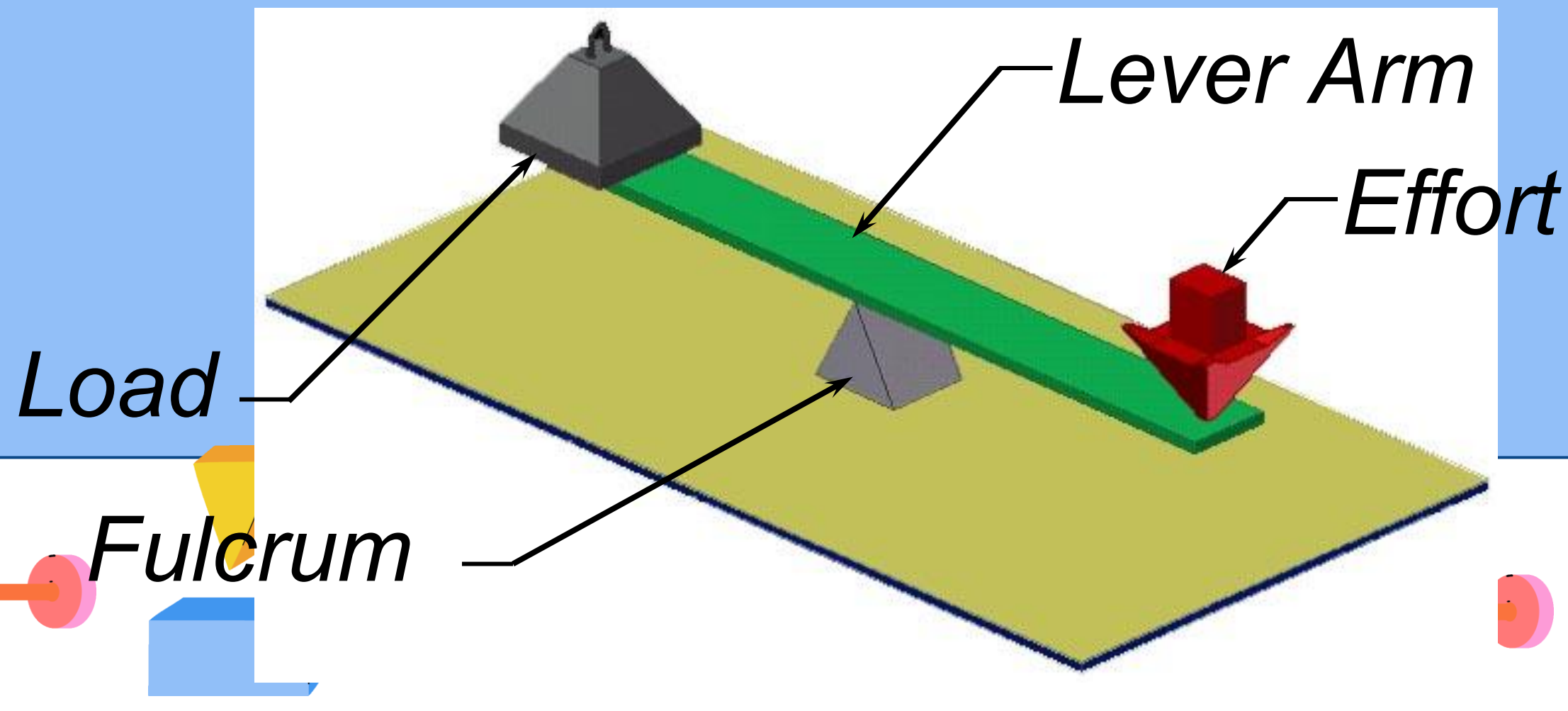
# LEVER

- It is a bar or board that rests on a support called fulcrum.

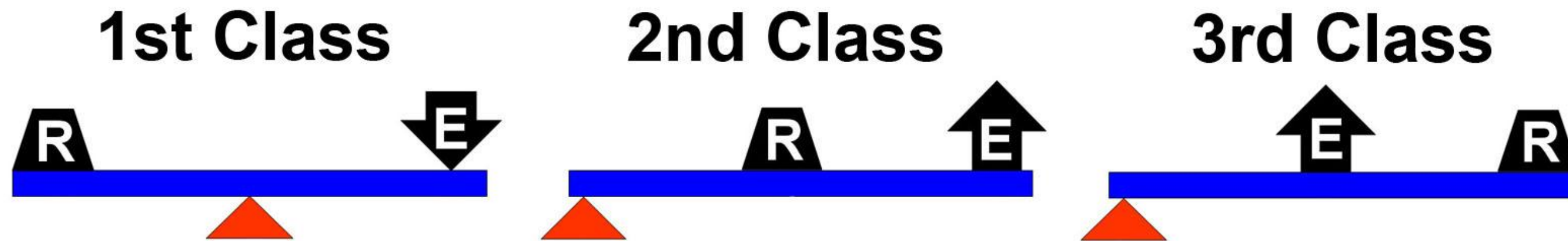


# Parts of a Lever

If the **fulcrum** is located at the midpoint of the **lever arm** and the **effort** is equal to the **load**, then the lever will be perfectly balanced.

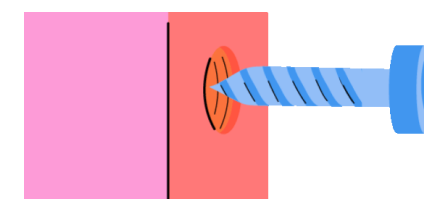
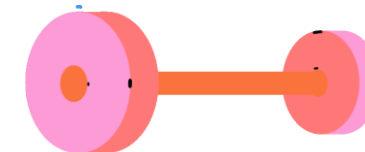
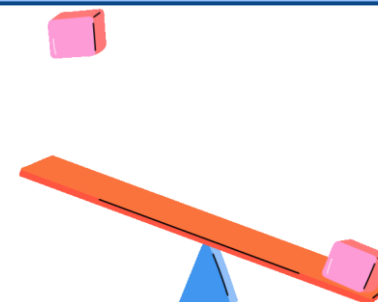
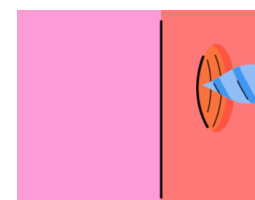
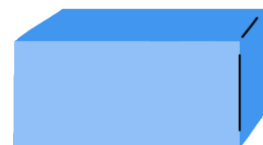
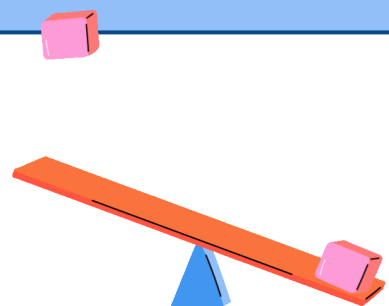
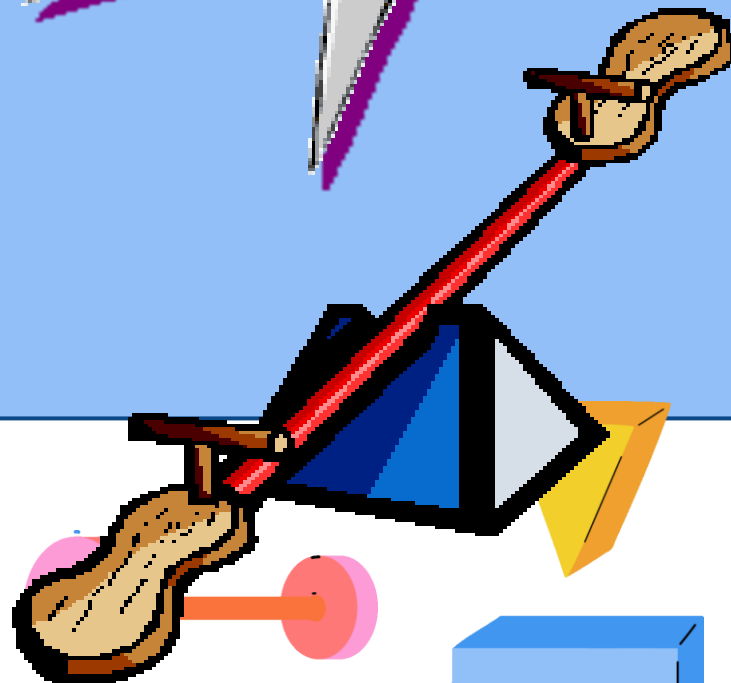
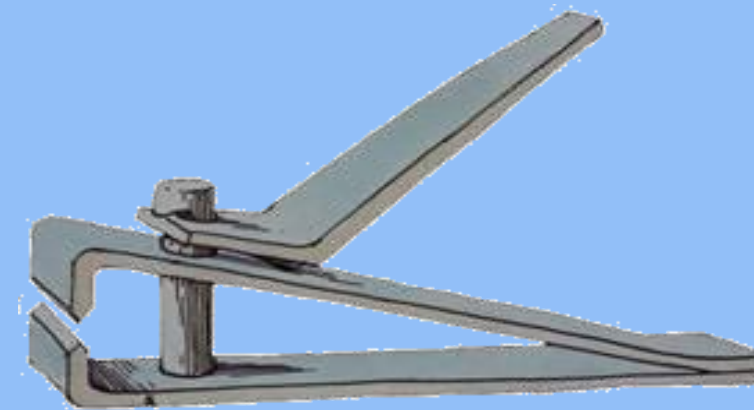
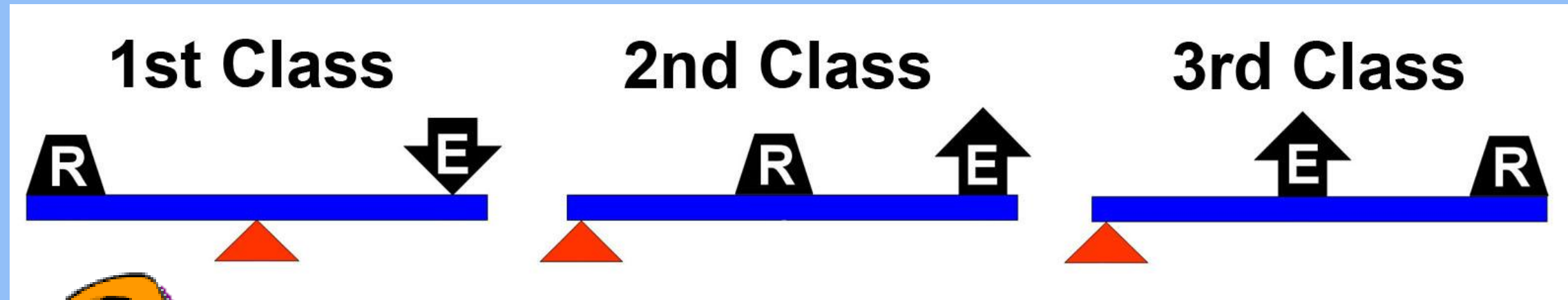


# Types of Levers



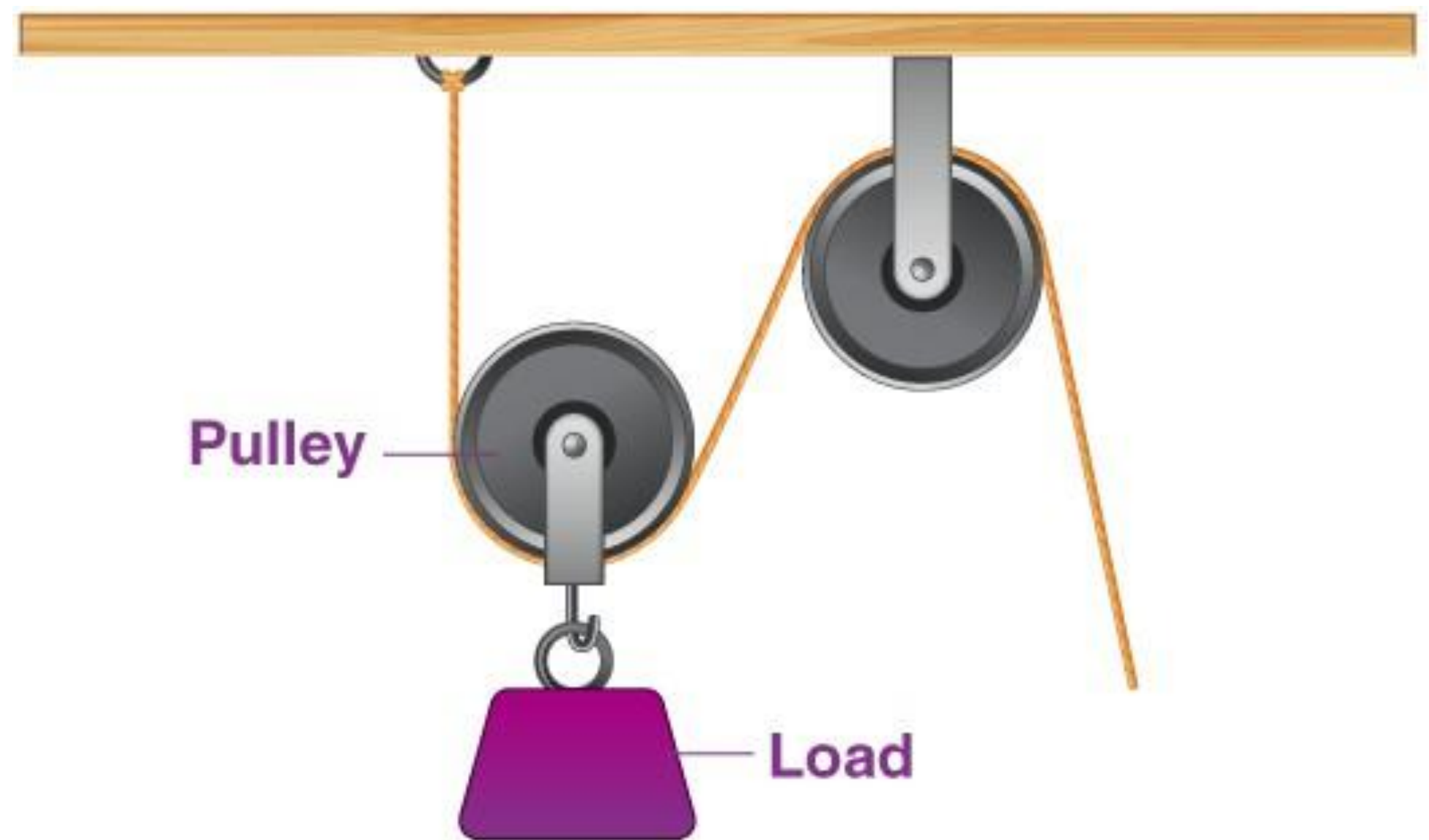
You can identify the different types of levers by the middle component.

# Examples of Levers

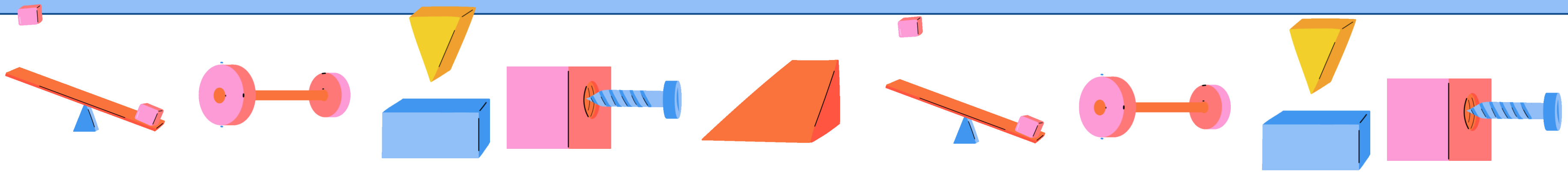
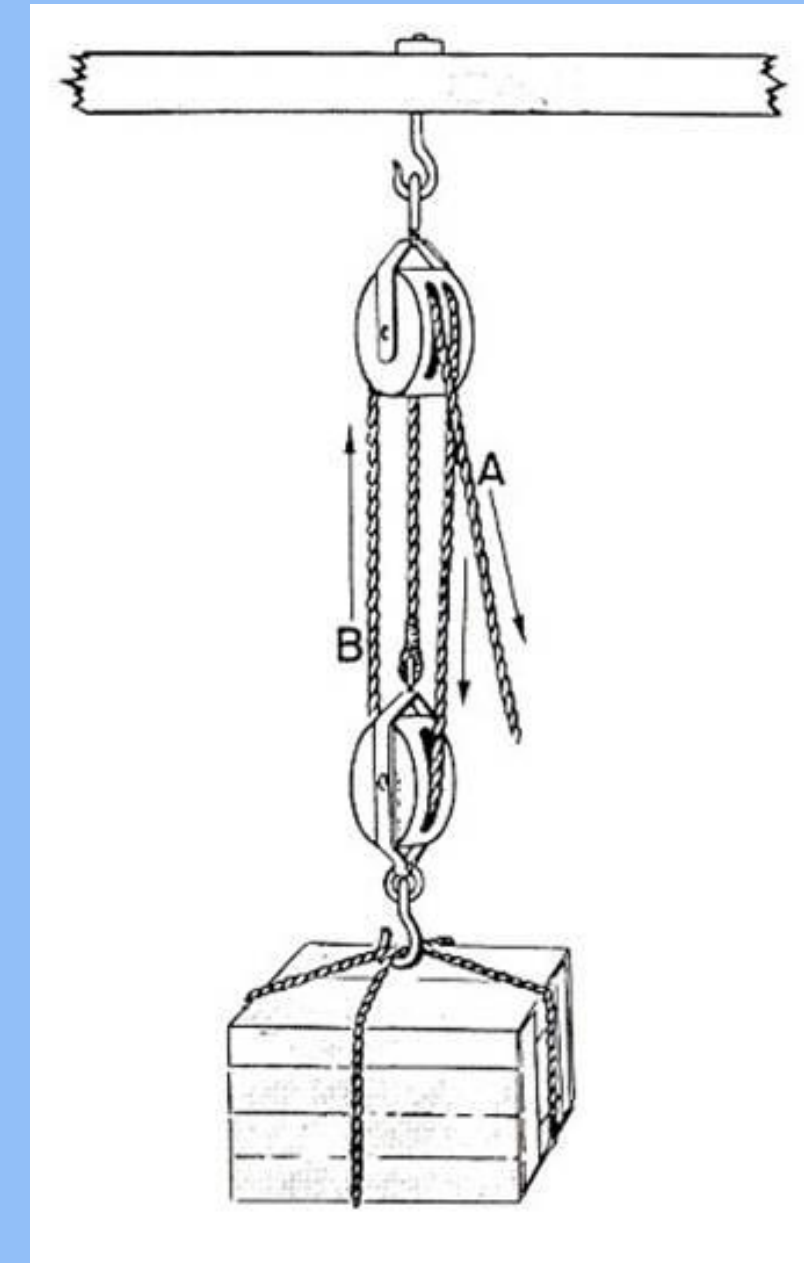
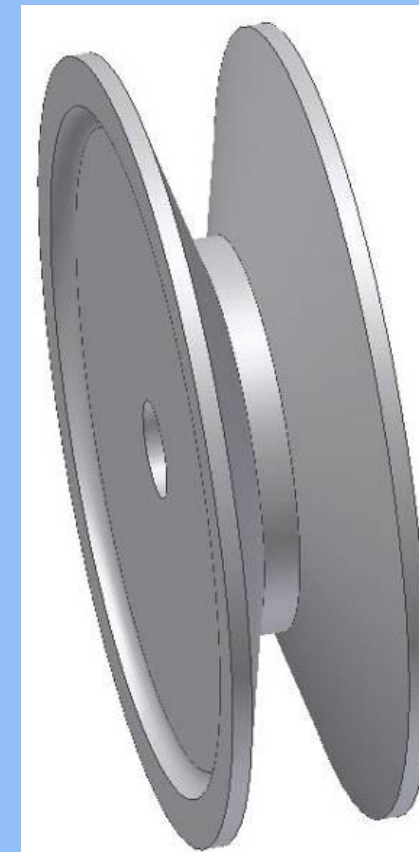


# PULLEY

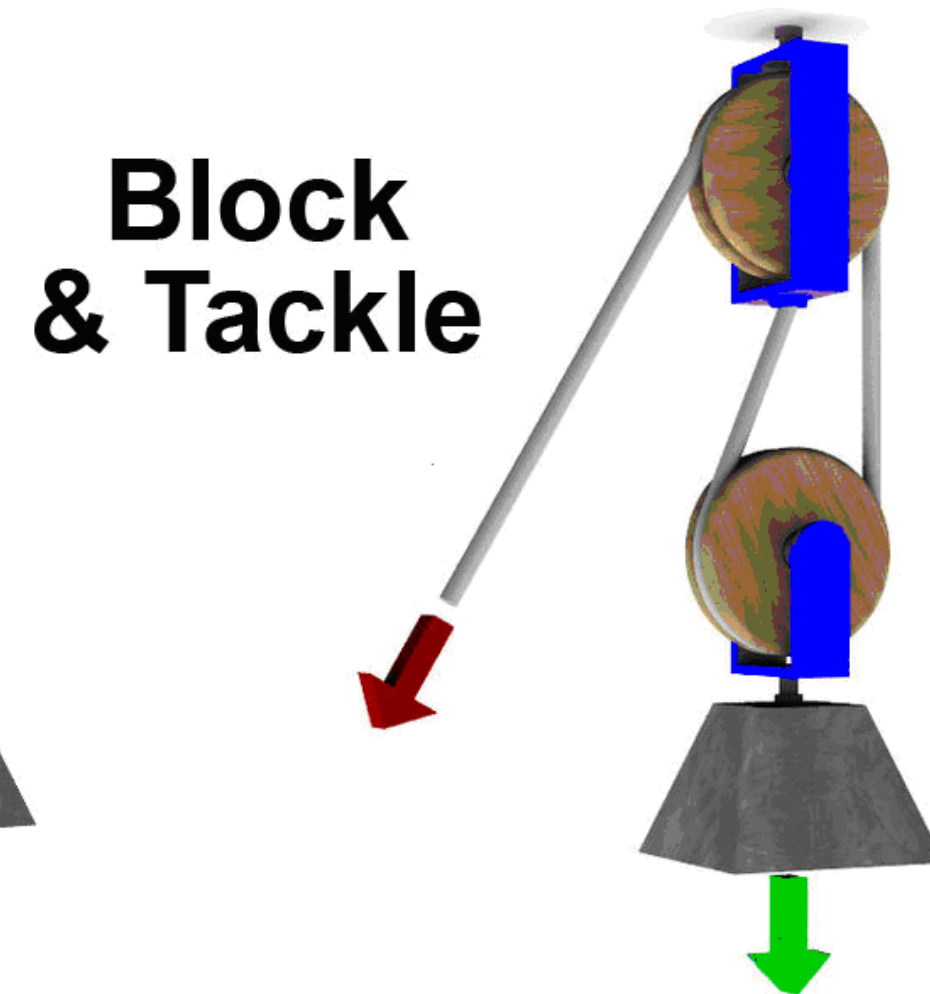
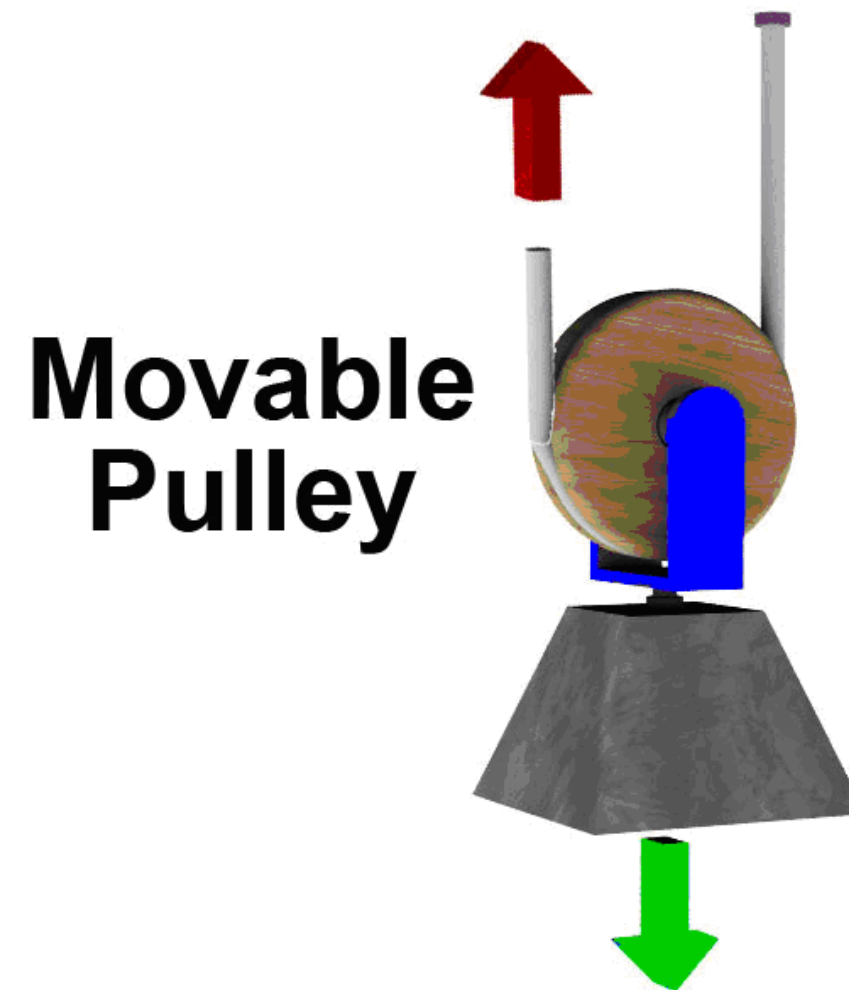
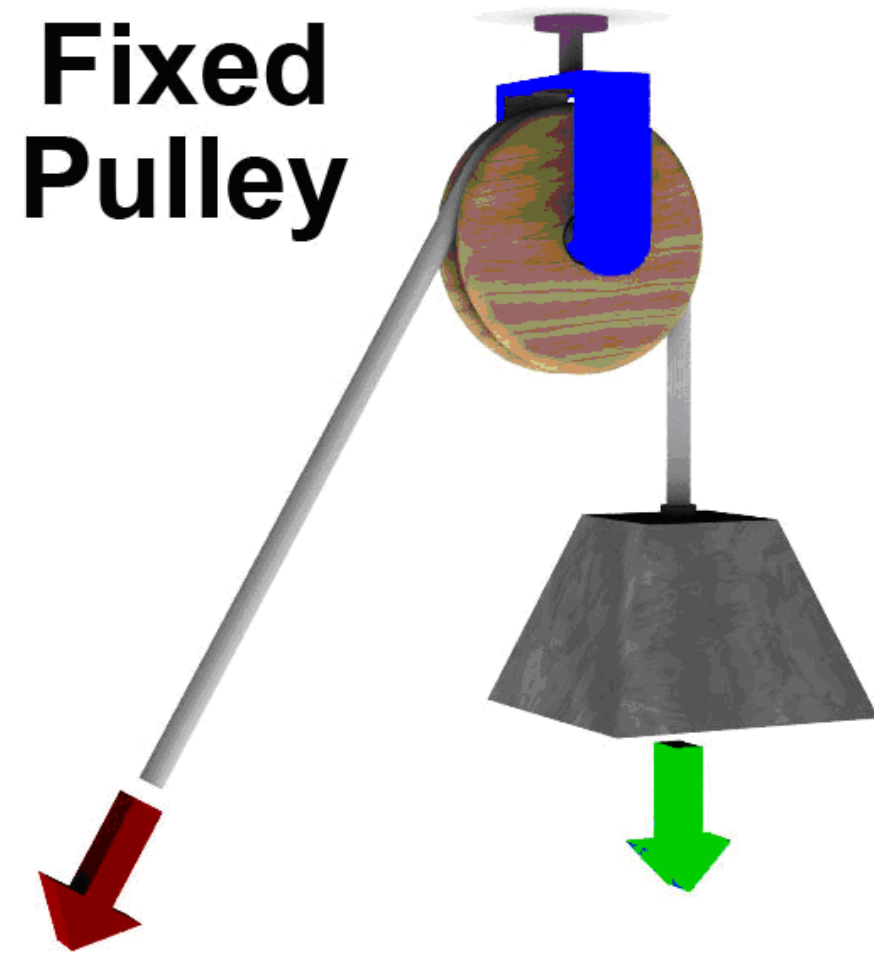
- It is a wheel that carries a flexible rope, cord, cable, chain, or belt on its rim.



# Examples of Pulleys

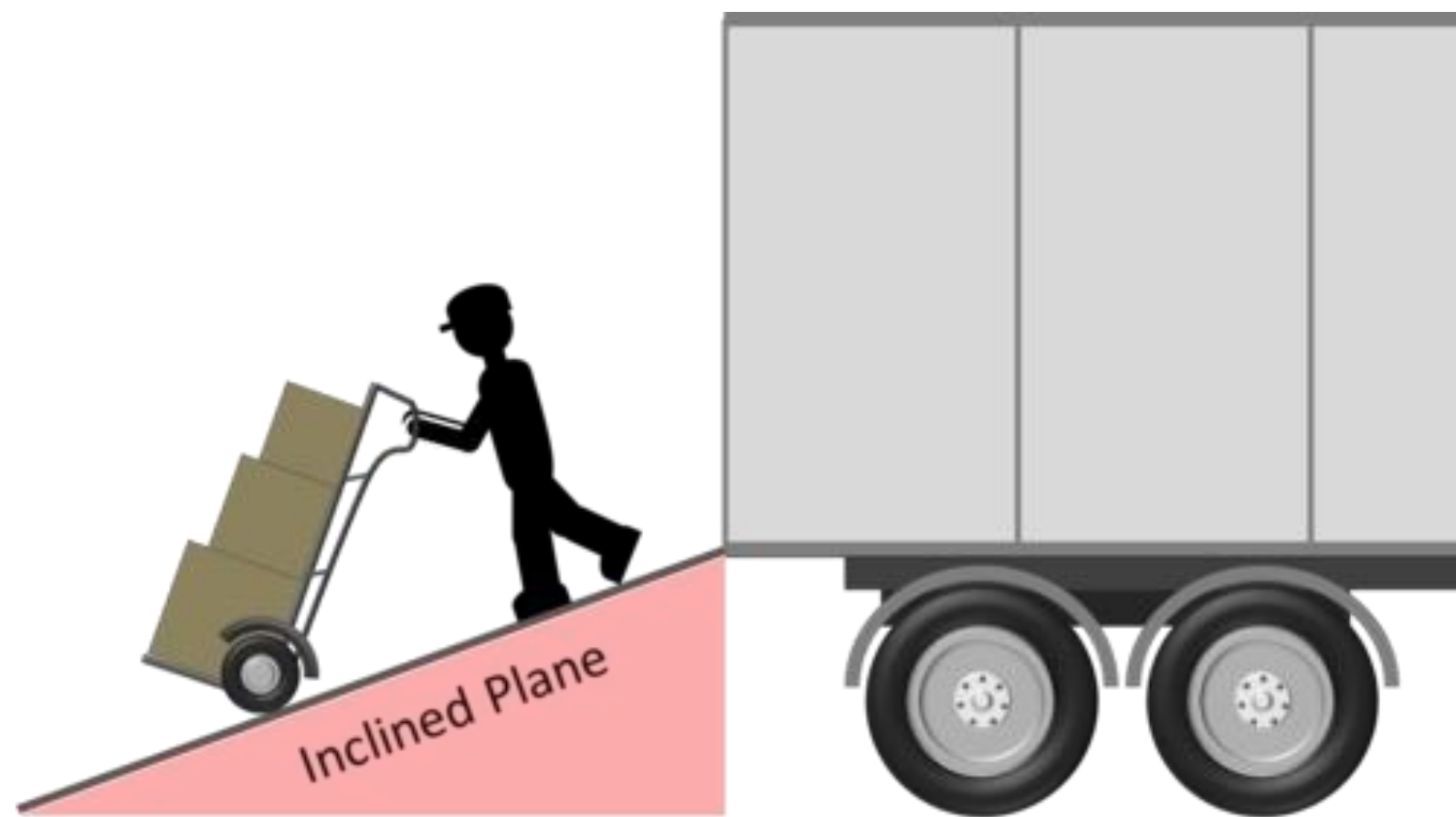


# Pulley Systems

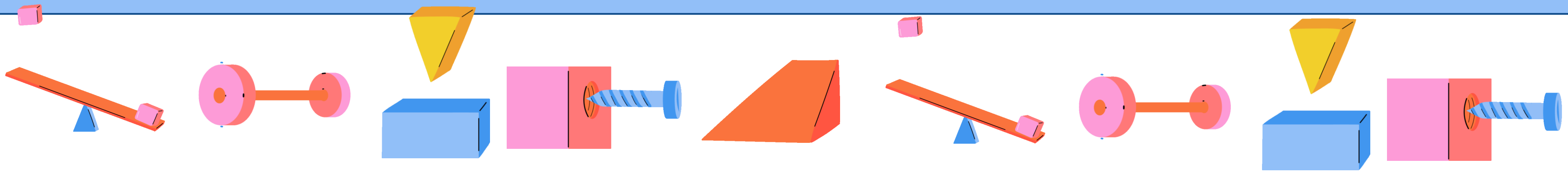
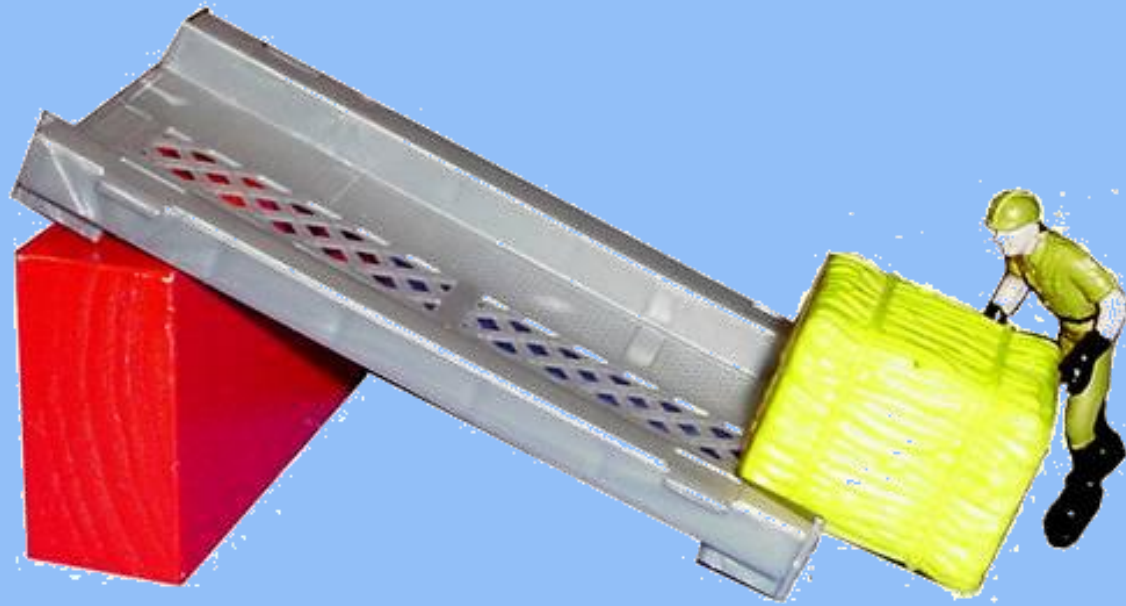


# INCLINED PLANE

- It consists of a sloping surface, used for raising heavy bodies.



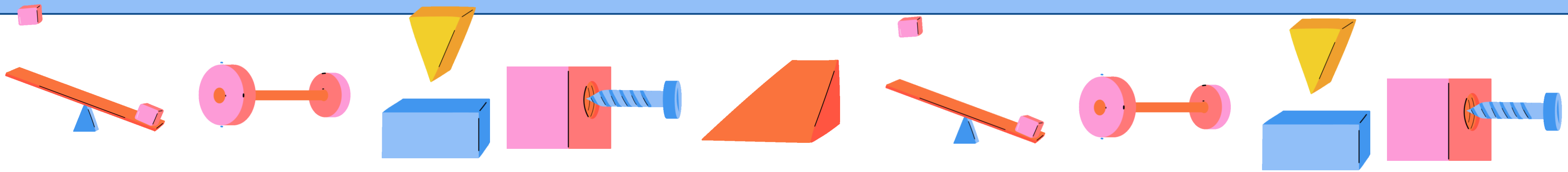
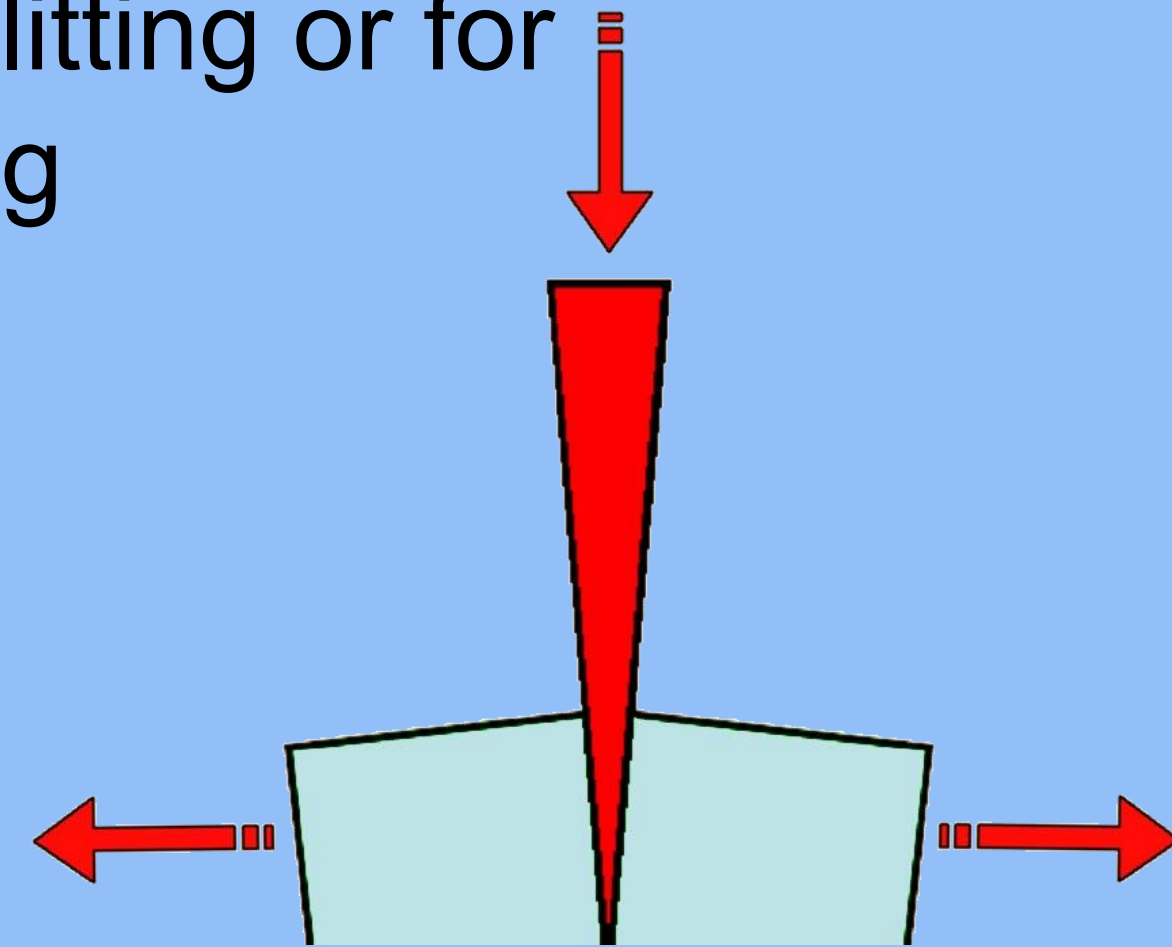
# Examples of Inclined Planes



# Wedge

# Functions as a *moving* inclined plane

Tapers to a thin edge and is used for splitting or for tightening by being driven into something



# WEDGE

- It is an object that tapers to a thin edge.



**Axe**



**Knife**



**Chisel**

# Examples of Wedges



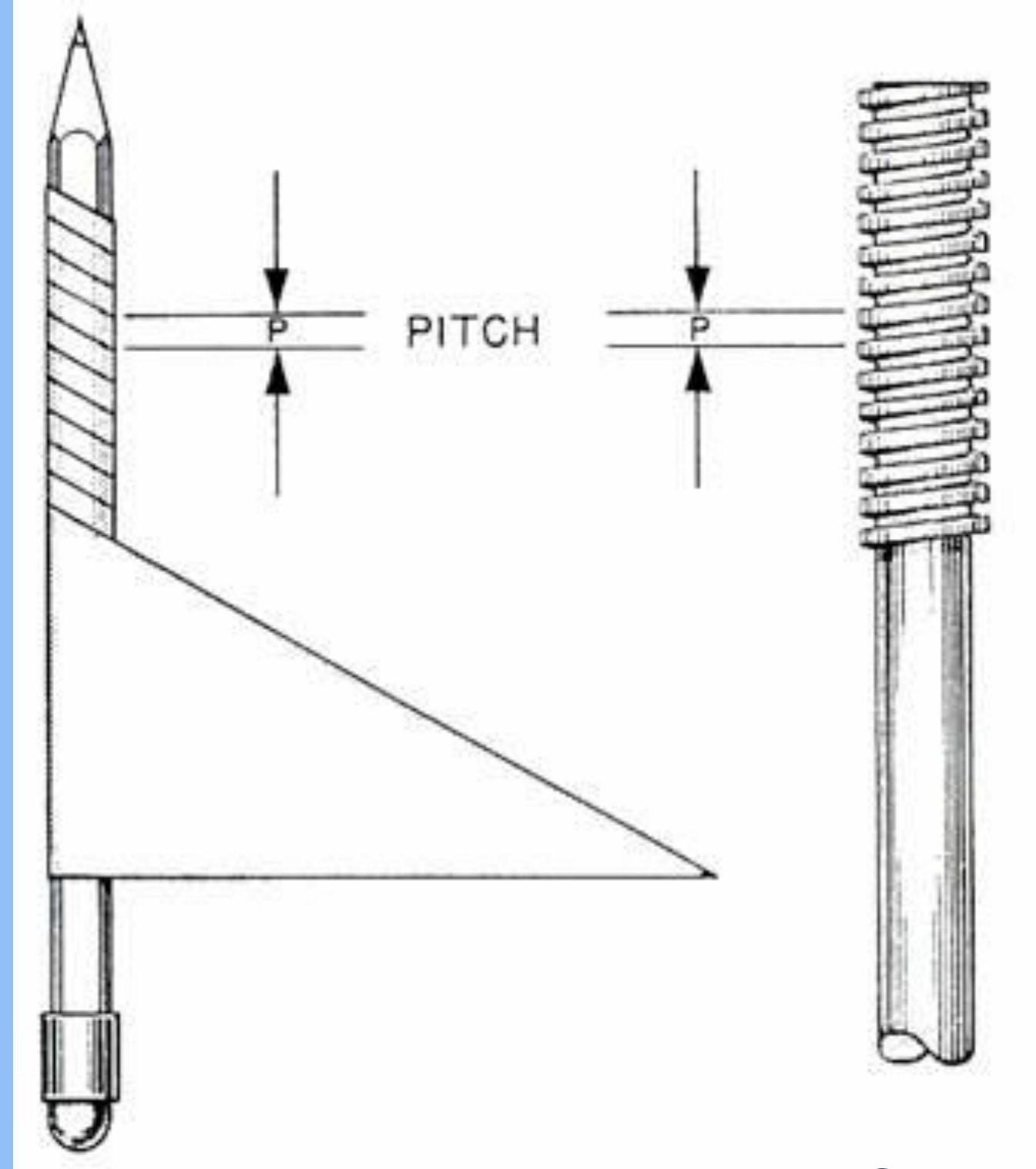
*Almost all cutting tools make use of the wedge.*



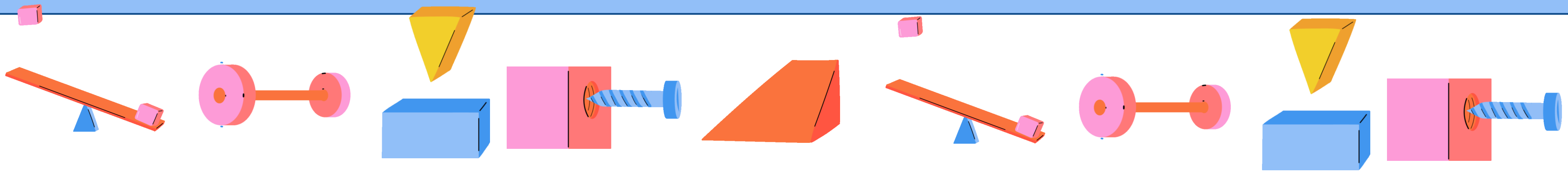
# Screw

An inclined plane wrapped around a cylinder, forming the path and pitch

Pitch is the distance between two threads

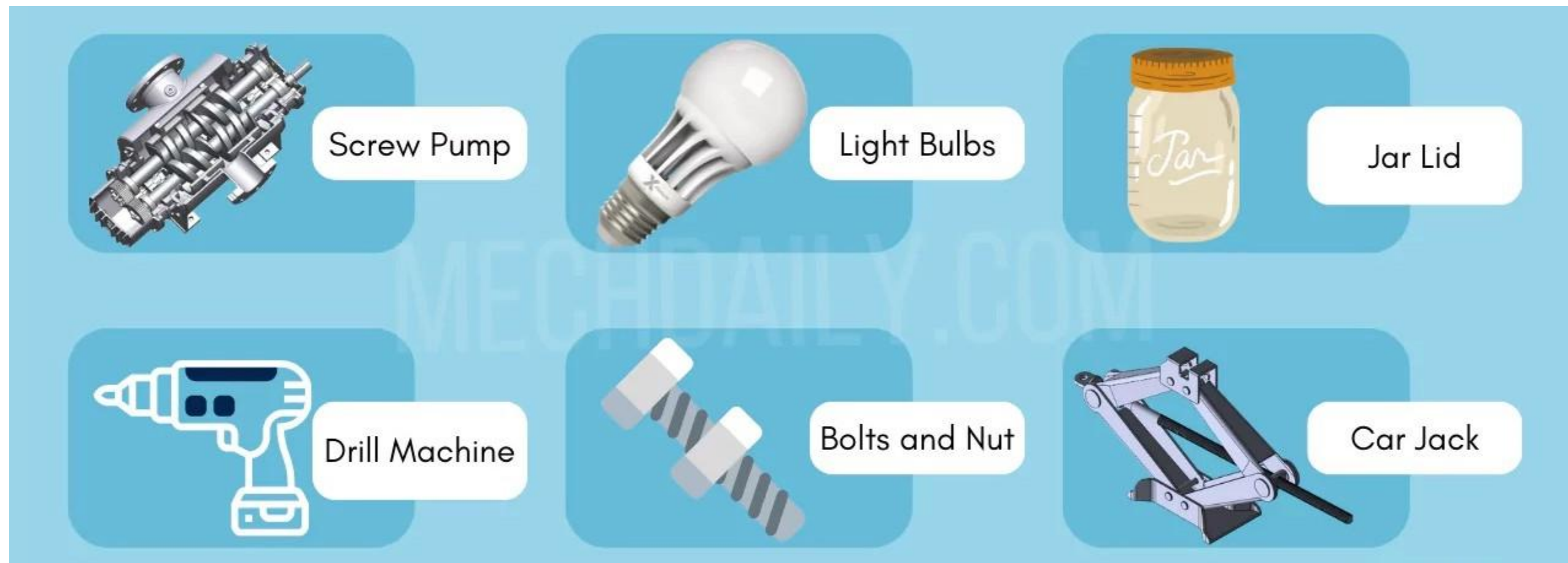


A screw is used as a threaded fastener.  
It changes rotary motion to linear force.



# SCREW

- It is usually cylindrical with a continuous helical rib.

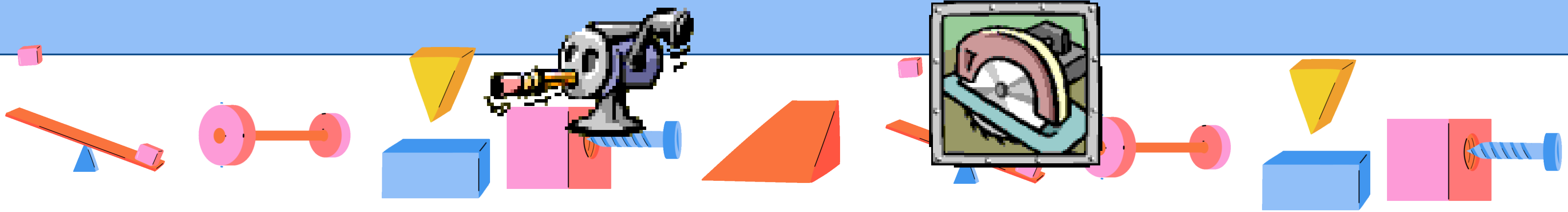


# Examples of Screws





The *wheel and axle* move together as a simple lever to lift or to move an item by rolling.



# WHEEL AND AXLE

- It is made up of a circular frame (the wheel) that revolves on a shaft or rod (the axle).

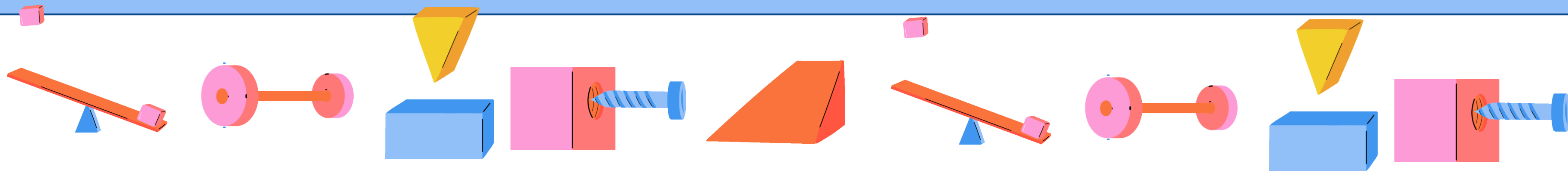
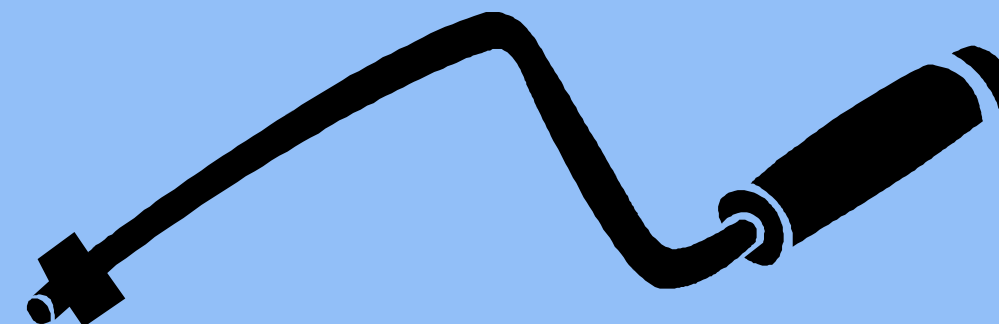
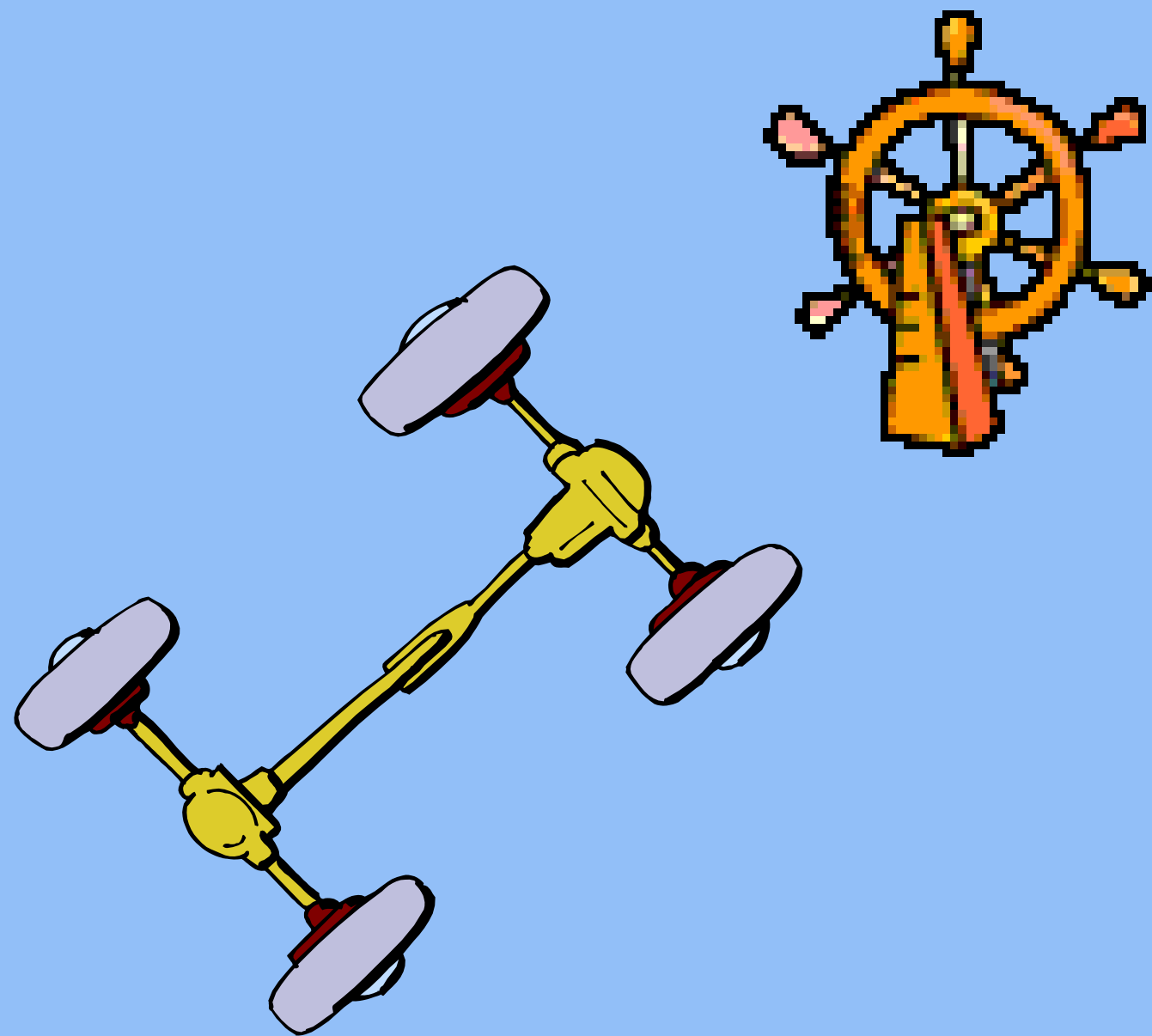


Bicycle



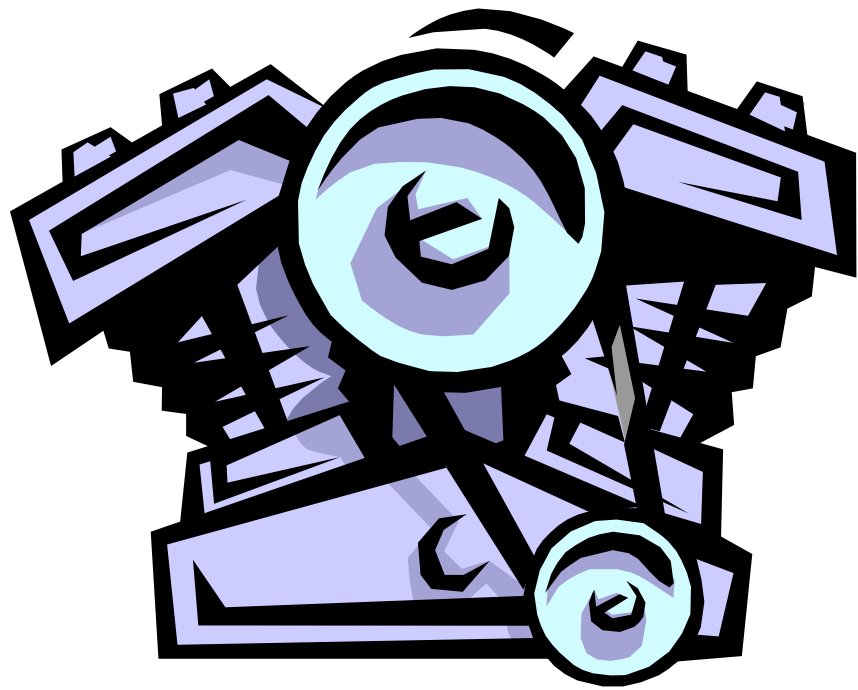
Ferris Wheel

# Examples of Wheel & Axles



# What Kind of Simple Machine Is This?

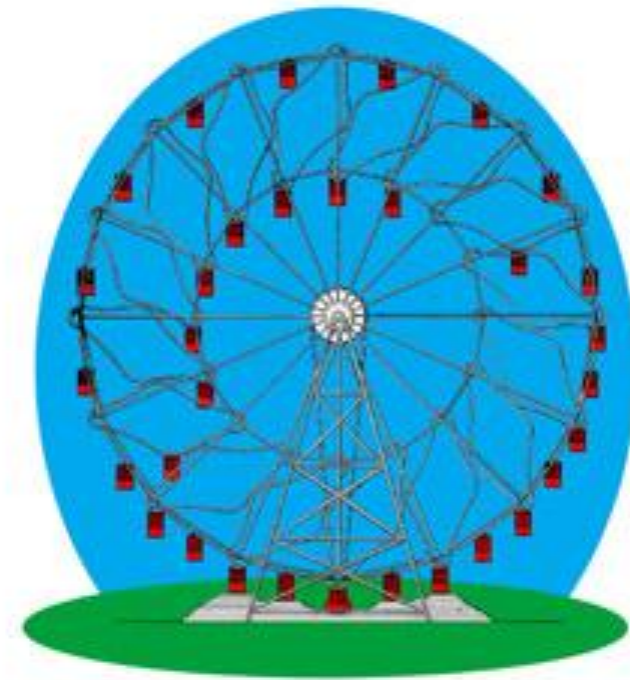
**Pulley**



**Wedge**



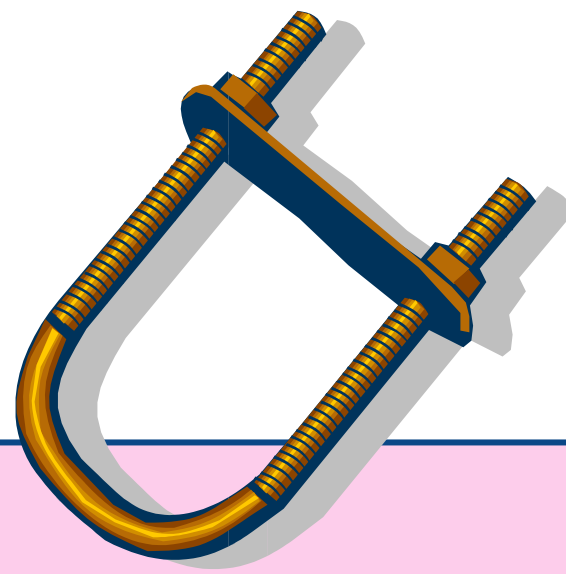
**Wheel and Axle**



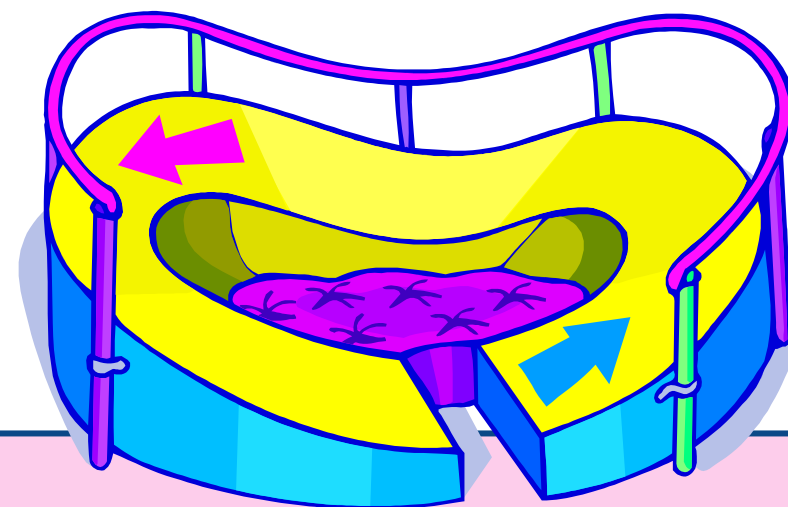
**Lever**



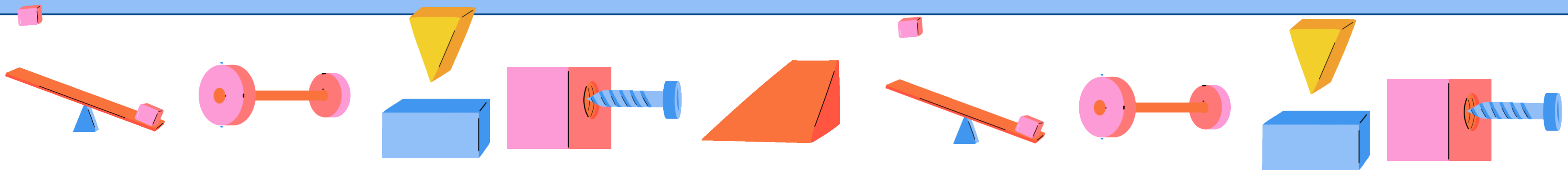
**Screw**



**Inclined Plane**



# COMPOUND MACHINES



# COMPOUND MACHINE

- A compound machine consists of multiple simple machines working together.
- When simple machines are combined, they create a complex tool capable of achieving greater mechanical advantage.

# COMPOUND MACHINE EXAMPLES

- Scissors
- Wheelbarrow
- Can opener
- Stapler
- Fishing rod
- Escalator
- Crane
- Clipper
- Fans
- Forklifts
- Conveyor belts

# Compound Machines

Compound machines consist of two or more simple machines put together.

Most devices are compound machines.

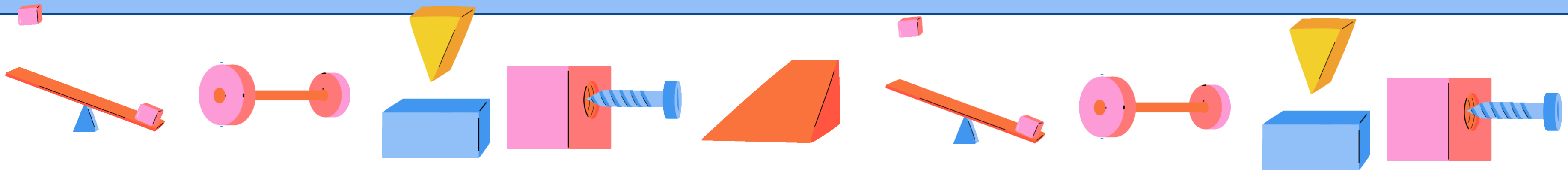
They can do more difficult jobs because their mechanical advantage is greater.



What two simple machines are in scissors?

First Class Lever and Wedge

# CHARACTERISTICS OF EFFICIENT MACHINES



# CHARACTERISTICS

## **1. High Mechanical Advantage**

- Reduces effort needed
- Example: scissors, pliers, gears in bicycles

# CHARACTERISTICS

## 1. High Mechanical Advantage



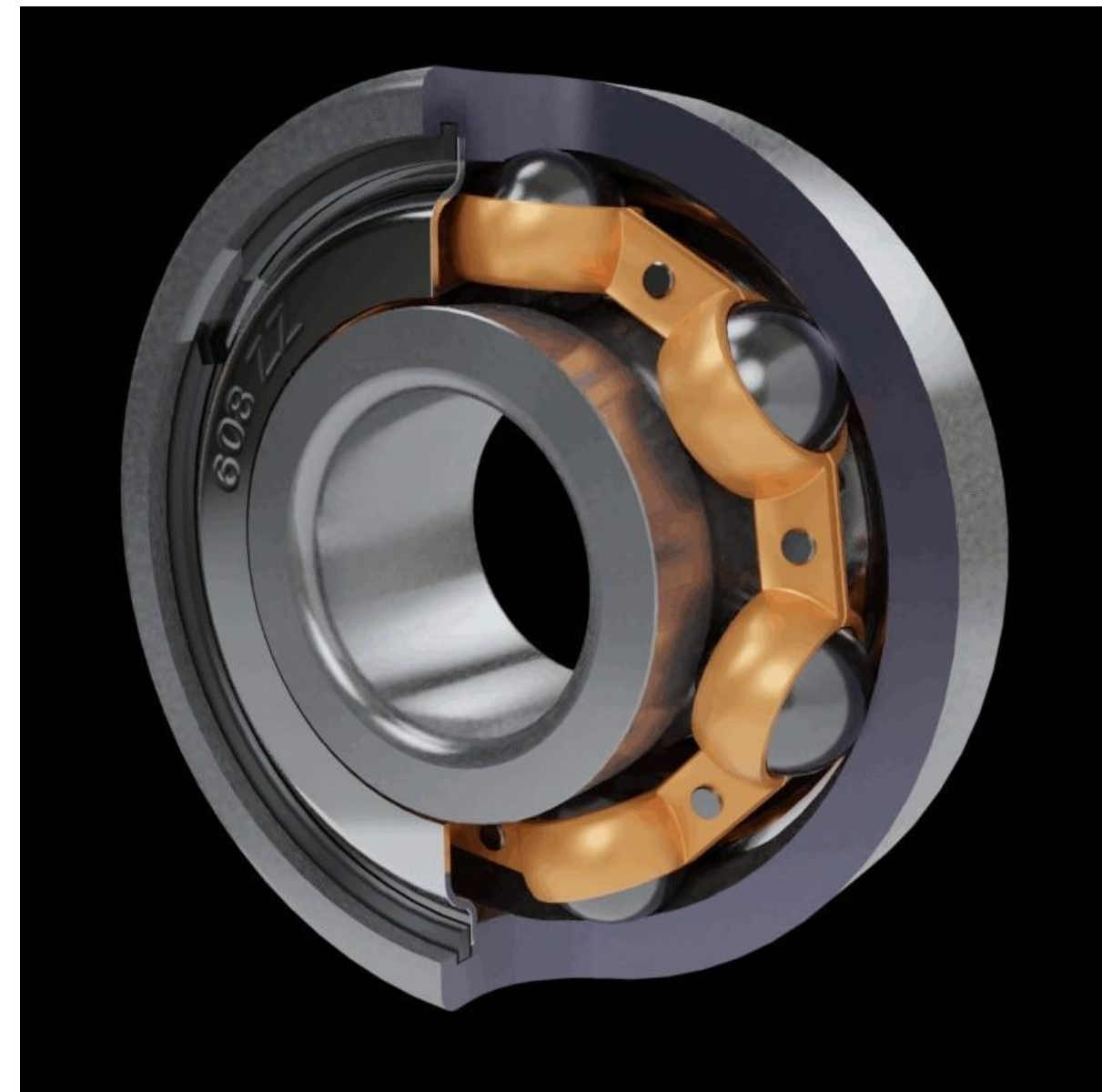
# CHARACTERISTICS

## **2. Low Energy Loss**

- Minimizes friction or heat
- Example: Maglev trains, heat pumps

# CHARACTERISTICS

## 2. Low Energy Loss



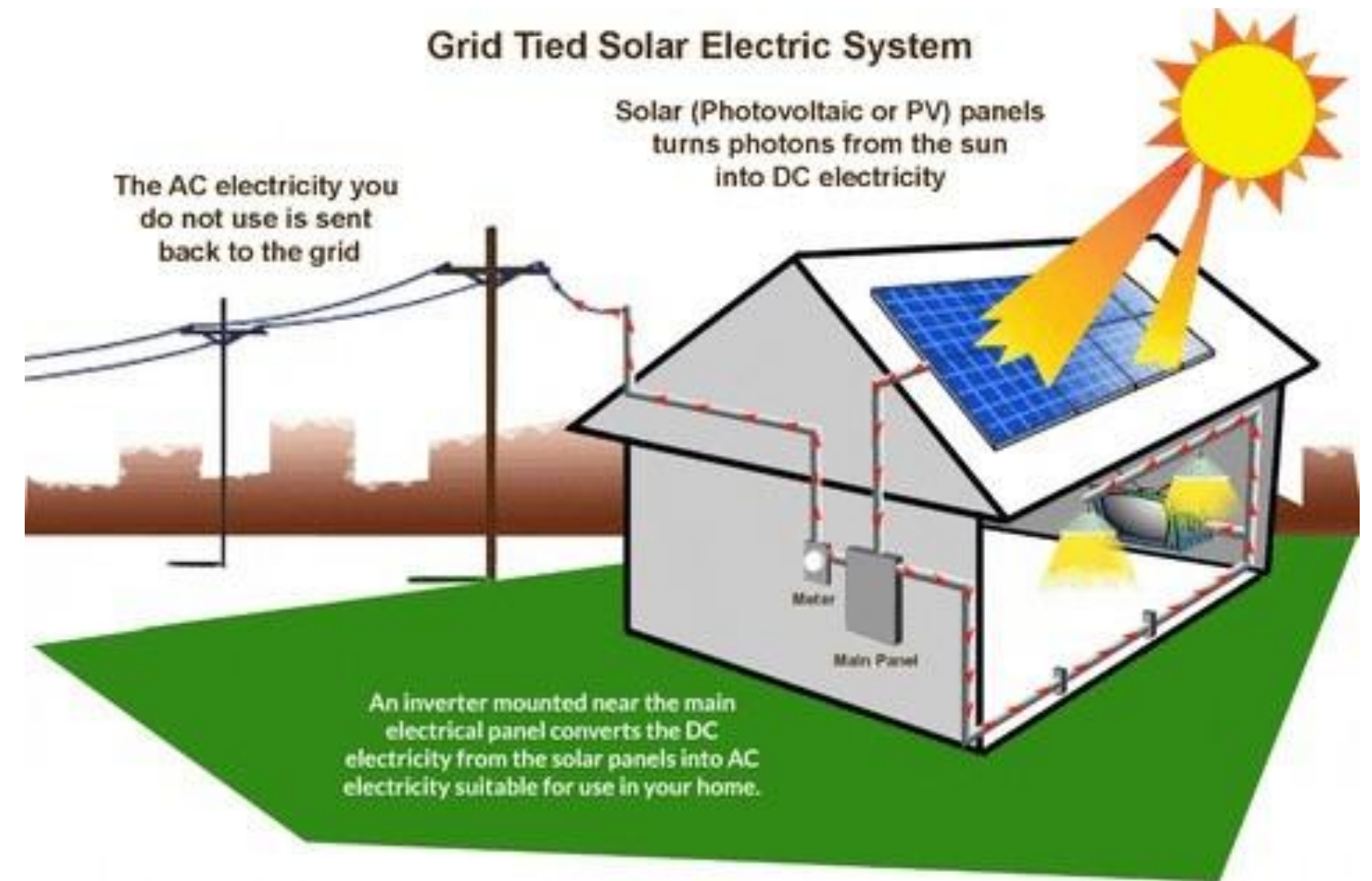
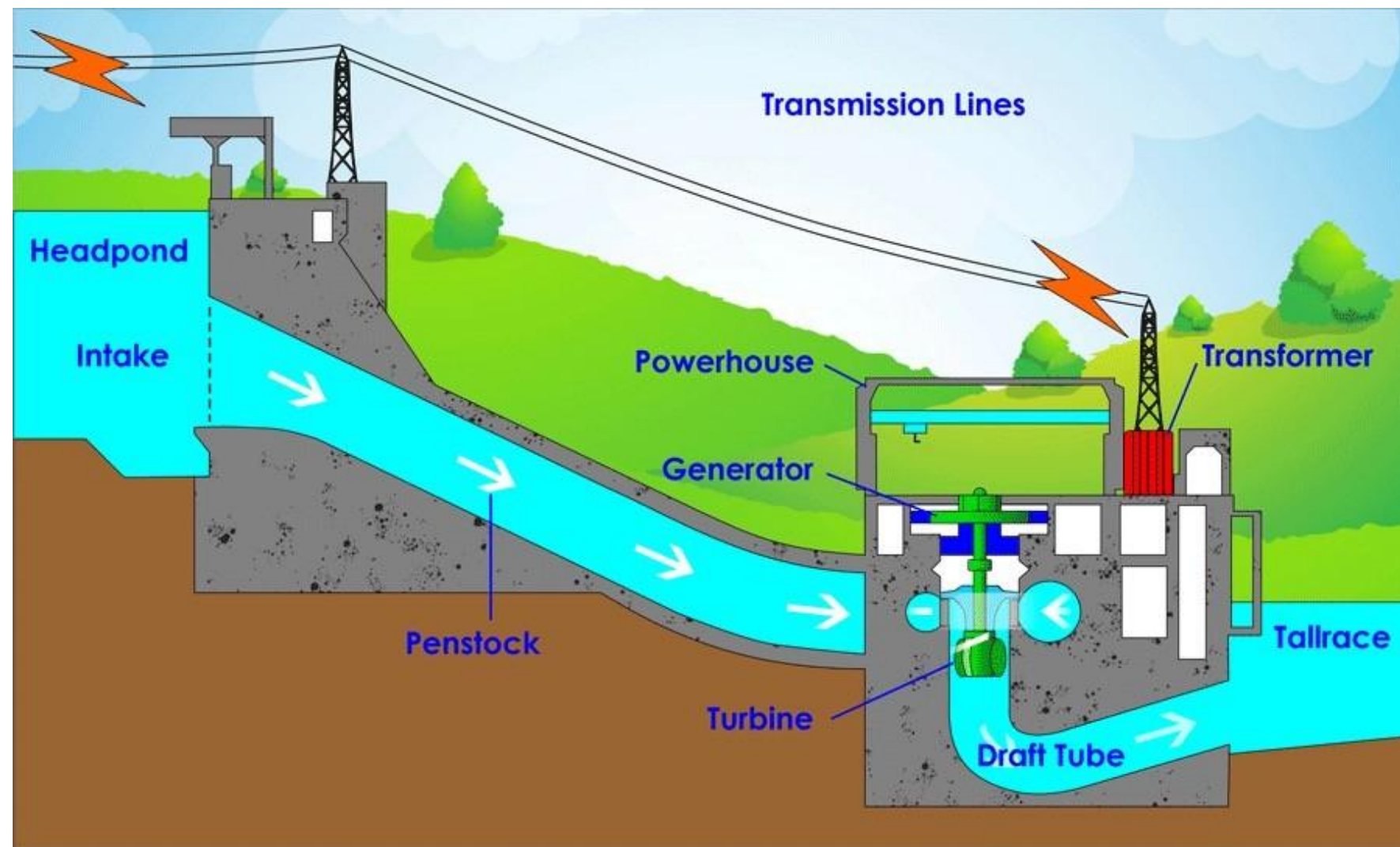
# CHARACTERISTICS

## **3. Effective Energy Conversion**

- Converts input to useful work
- Example: blender, battery-powered tools

# CHARACTERISTICS

## 3. Effective Energy Conversion



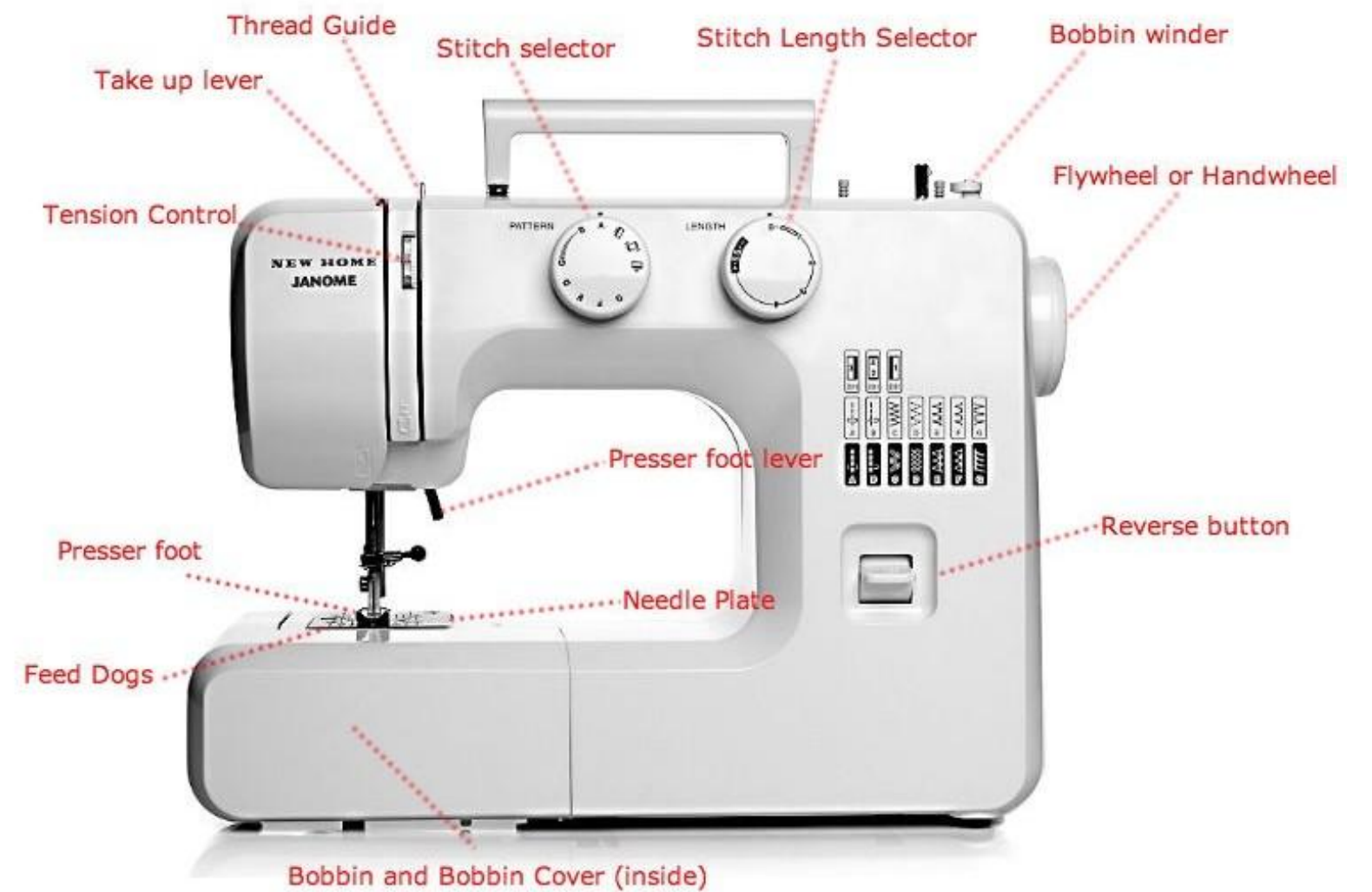
# CHARACTERISTICS

## **4. Smart Machine Combination**

- Effective use of simple machines
- Example: Bicycles, can openers

# CHARACTERISTICS

## 4. Smart Machine Combination



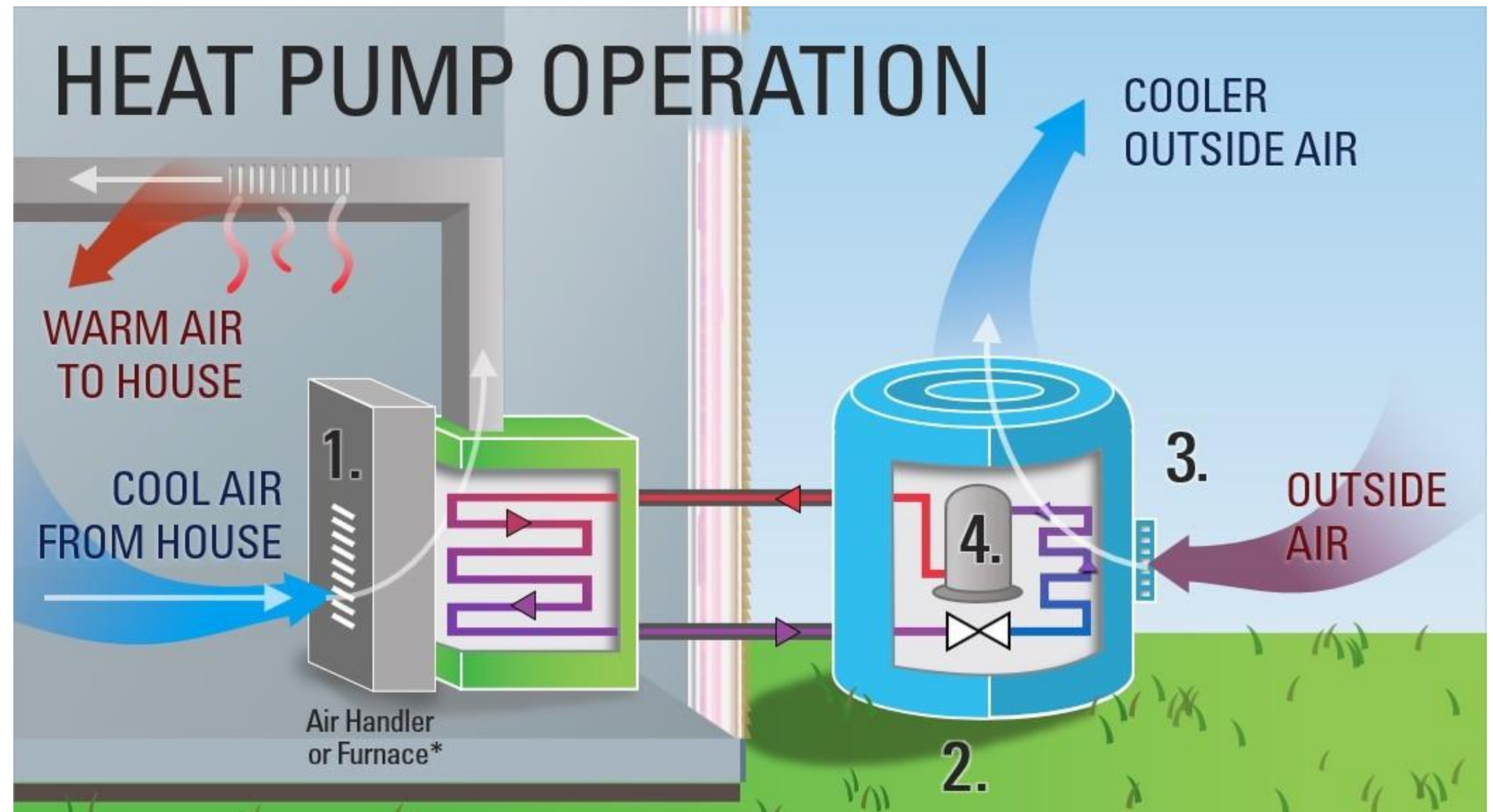
# CHARACTERISTICS

## 5. High Efficiency Ratio

- $\text{Output over Input Work} \times 100\%$
- Example: LED lights, electric vehicles

# CHARACTERISTICS

## 5. High Efficiency Ratio



# CHARACTERISTICS

## **6. Easy to Use and Maintain**

- Long-lasting and safe
- Example: microwave oven, electric kettles

# CHARACTERISTICS

## 6. Easy to Use and Maintain



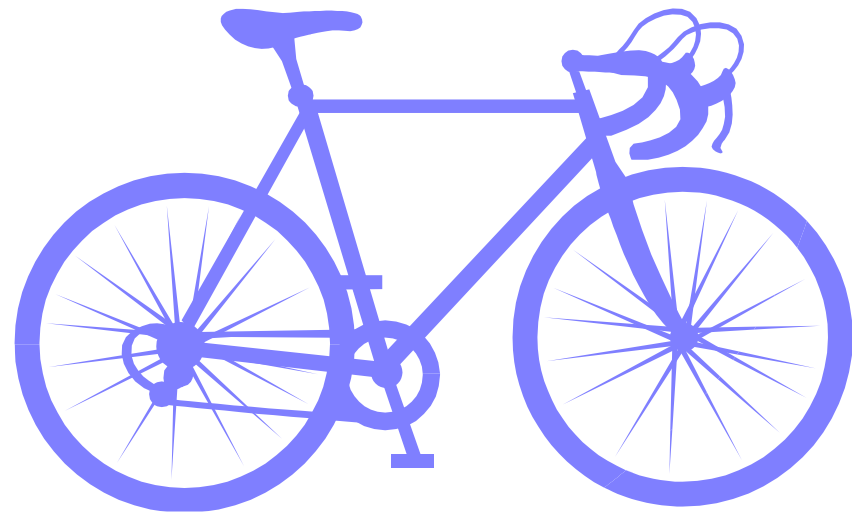
# SUMMARY

**Simple machines** are basic tools like levers, pulleys, inclined planes, wheels and axles, screws, and wedges that make work easier. When two or more simple machines are combined, they form **compound machines**, such as scissors or bicycles, which perform more complex tasks efficiently. An **efficient machine** maximizes output while minimizing energy loss, characterized by high mechanical advantage, low friction, durability, smooth operation, and effective energy use.

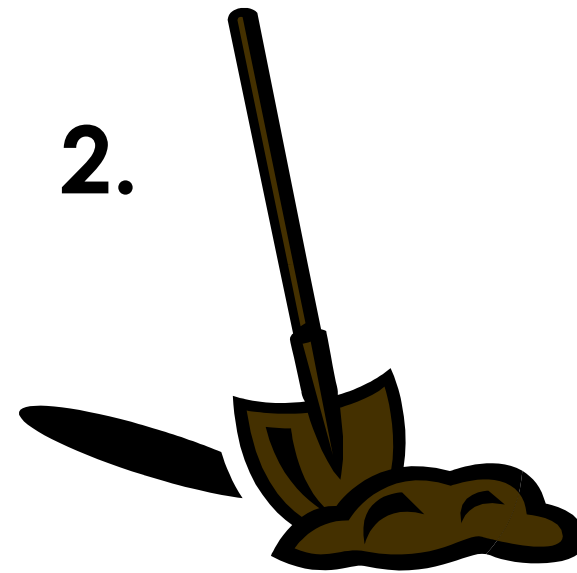
# ASSESSMENT

## Name the Simple Machines present in Each Compound Machine

1.



2.



3.



4.



5.



| Questions                     |                      | Choices     |
|-------------------------------|----------------------|-------------|
| 1. Lever                      | <input type="text"/> | Scissors    |
| 2. Wheel and axle             | <input type="text"/> | Rolling pin |
| 3. Pulley                     | <input type="text"/> | Axe head    |
| 4. Inclined plane             | <input type="text"/> | Drill bit   |
| 5. Wedge                      | <input type="text"/> | Escalator   |
| 6. Screw                      | <input type="text"/> | Flagpole    |
| 7. Wedge and lever            | <input type="text"/> | Crowbar     |
| 8. Lever and wheel and axle   | <input type="text"/> | Wheelbarrow |
| 9. Pulleys and inclined plane | <input type="text"/> | Ramp        |
| 10. Screw and lever           | <input type="text"/> | Corkscrew   |

# Build a Working Simple Machine

- ☐ Choose **one** of the six simple machines:
  - Lever
  - Wheel and Axle
  - Pulley
  - Inclined Plane
  - Wedge
  - Screw
- ☐ Research how your chosen simple machine works and identify its everyday applications.
- ☐ Design and build a **working model** that clearly demonstrates the function of your chosen simple machine.
- ☐ Use **recycled, inexpensive, or locally available materials** whenever possible.
- ☐ Ensure that your model is **safe, durable, neat, and functional**.
- ☐ Prepare a **5–7 minute presentation** demonstrating your simple machine.

# During the Presentation

Each group must:

- Introduce the chosen simple machine.
- Identify its main parts.
- Explain how it works.
- Demonstrate the model in action.
- Discuss at least **three real-life applications** of the simple machine.
- Explain how the machine reduces effort or makes work easier.

# Submission Requirements

Submit the following on the presentation day:

- Working model of the simple machine
- One-page explanation containing:
  - ✓ Name of the simple machine
  - ✓ Members of the group
  - ✓ Materials used
  - ✓ Procedure in constructing the model
  - ✓ Principle of operation
  - ✓ Real-life applications
  - ✓ Attached the criteria

| Criteria                           | Points     |
|------------------------------------|------------|
| Functionality of the Model         | <b>30</b>  |
| Creativity and Design              | <b>20</b>  |
| Accuracy of Scientific Explanation | <b>20</b>  |
| Presentation and Demonstration     | <b>20</b>  |
| Teamwork and Organization          | <b>10</b>  |
| <b>Total</b>                       | <b>100</b> |

## Reminders

- Every group member must actively participate in the construction and presentation.
- Practice your demonstration before presentation day.
- Prioritize safety when using tools and materials.
- Submit your project on time. Late submissions may receive a deduction based on the teacher's policy.

DIY SIMPLE | COMPOUND  
MACHINE PROJECT

TikTok  
© steph\_mommy.co



Naya Isabelle A. Catayaban  
Science Project - Simple Machine

Watch how this simple bottle  
turns into a real Water machine 🤖

TikTok  
@ molly\_ b



# INDEPENDENT RECAP ACTIVITY

Visit the link on

[Simple and Compound Machines Quiz |  
Technology and Computers | 10 Questions](#)